

Fonts and Alignment Filter Demonstration

Nandakumar Chandrasekhar

2026-05-22

This document demonstrates every feature provided by the `fonts-and-alignment` Lua filter. The syntax shown in the code blocks directly produces the output below them, and the same Markdown source produces faithful output in both LaTeX/PDF and HTML.

1. Font Sizing Scale (Inline)

Nine sizing hooks scale relative to the document's base font size.

Size	Syntax	Output
tiny	<code>[Sample]{.pfa-text-3xs}</code>	Sample
scriptsize	<code>[Sample]{.pfa-text-2xs}</code>	Sample
footnotesize	<code>[Sample]{.pfa-text-xs}</code>	Sample
small	<code>[Sample]{.pfa-text-sm}</code>	Sample
normal	<code>[Sample]{.pfa-text-normal}</code>	Sample
large	<code>[Sample]{.pfa-text-lg}</code>	Sample
Large	<code>[Sample]{.pfa-text-xl}</code>	Sample
LARGE	<code>[Sample]{.pfa-text-2xl}</code>	Sample
huge	<code>[Sample]{.pfa-text-3xl}</code>	Sample

2. Font Sizing Scale (Block)

The same sizing classes apply to fenced div blocks. This is useful for sidebars, captions, or callouts.

```
::: pfa-text-sm
This entire paragraph renders at the `small` size. Block-level sizing ensures
↪ the content consistently shrinks or grows as a unified component.
:::
```

This entire paragraph renders at the `small` size. Block-level sizing ensures the content consistently shrinks or grows as a unified component.

```
::: pfa-text-lg
This entire paragraph renders at the `large` size — well suited for pull quotes
↪ or emphasized opening text.
:::
```

This entire paragraph renders at the `large` size — well suited for pull quotes or emphasized opening text.

3. Font Weights, Shapes, and Families (Inline)

Every supported typographic style is available as an inline span constraint.

Style	Syntax	Output
Bold	<code>[Sample]{.pfa-font-bold}</code>	Sample
Medium	<code>[Sample]{.pfa-font-medium}</code>	Sample
Italic	<code>[Sample]{.pfa-font-italic}</code>	<i>Sample</i>
Slanted	<code>[Sample]{.pfa-font-slanted}</code>	<i>Sample</i>
Upright	<code>[Sample]{.pfa-font-upright}</code>	Sample
Emphasis	<code>[Sample]{.pfa-font-emphasis}</code>	<i>Sample</i>
Serif	<code>[Sample]{.pfa-font-serif}</code>	Sample
Sans-Serif	<code>[Sample]{.pfa-font-sans}</code>	Sample
Monospace	<code>[Sample]{.pfa-font-mono}</code>	Sample
Small Caps	<code>[Sample]{.pfa-font-smallcaps}</code>	SAMPLE
Normal Reset	<code>[Sample]{.pfa-font-normal}</code>	Sample

4. Font Weights, Shapes, and Families (Block)

The same classes work at the block level, applying the typographic voice across all internal sentences without requiring inline markers.

```
::: pfa-font-sans
This paragraph is rendered using the document's sans-serif family. Block-scoped
↪ family switches are convenient for code blocks or structural sections.
:::
```

```

::: pfa-font-sans
This paragraph is rendered using the document's sans-serif family. Block-scoped
↪ family switches are convenient for code blocks or structural sections.
:::

```markdown
::: pfa-font-smallcaps
Section headings or short callouts can be rendered entirely in small caps using
↪ the block form.
:::

```

SECTION HEADINGS OR SHORT CALLOUTS CAN BE RENDERED ENTIRELY IN SMALL CAPS USING THE BLOCK FORM.

## 5. Text Casing Transformations

Casing classes rewrite the underlying Abstract Syntax Tree (AST) text nodes rather than applying CSS formatting, meaning the transformation survives raw copy-paste operations.

### Inline Casing

Casing	Syntax	Output
Uppercase	[coerced]{.pfa-uppercase}	COERCED
Lowercase	[COERCED]{.pfa-lowercase}	coerced

### Block Casing

```

::: pfa-uppercase
this block is uniformly uppercased, even across paragraph boundaries.
:::

```

THIS BLOCK IS UNIFORMLY UPPERCASED, EVEN ACROSS PARAGRAPH BOUNDARIES.

## 6. Inline Text Decorations

Underline and strikeout variants are routed through the `ulem` LaTeX package and mapped to matching standard CSS.

Decoration	Syntax	Output
Single Underline	<code>[Sample]{.pfa-text-underline}</code>	<u>Sample</u>
Double Underline	<code>[Sample]{.pfa-text-underline-double}</code>	<u><u>Sample</u></u>
Dashed Underline	<code>[Sample]{.pfa-text-underline-dashed}</code>	<u>Sample</u>
Dotted Underline	<code>[Sample]{.pfa-text-underline-dotted}</code>	<u>Sample</u>
Wavy Underline	<code>[Sample]{.pfa-text-underline-wave}</code>	<u>Sample</u>
Strikeout	<code>[Sample]{.pfa-text-strikeout}</code>	<del>Sample</del>
Marked Out	<code>[Sample]{.pfa-text-markout}</code>	<del>Sample</del>

## 7. Color Resolution

The `pfa-color` attribute natively parses standard CSS3 named colors, six-digit hex codes, and three-digit hex shorthands.

Input Type	Syntax	Output
CSS Named	<code>[Sample]{pfa-color="crimson"}</code>	Sample
Hex Shorthand	<code>[Sample]{pfa-color="#333"}</code>	Sample
Hex Full	<code>[Sample]{pfa-color="#2E8B57"}</code>	Sample

### Flexible Spelling of Named Colors

The engine sanitizes inputs automatically. It is case-insensitive and ignores spaces, hyphens, and underscores. All syntaxes below successfully map to the standard `mediumvioletred`.

Variant	Syntax	Output
Lowercase	<code>[Color]{pfa-color="mediumvioletred"}</code>	Color
Title Case	<code>[Color]{pfa-color="MediumVioletRed"}</code>	Color
Spaces	<code>[Color]{pfa-color="Medium Violet Red"}</code>	Color
Hyphens	<code>[Color]{pfa-color="medium-violet-red"}</code>	Color

## Block-Level Color Constraints with Inline Overrides

Colors applied to a block automatically inherit downward, but nested spans can explicitly override the parent value.

```
::: {pfa-color="darkslategray"}
The parent block carries a default color of dark slate gray. [This inline span
↪ overrides the color to tomato red.]{pfa-color="tomato"} The surrounding
↪ text continues in the parent color until the block closes.
:::
```

The parent block carries a default color of dark slate gray. **This inline span overrides the color to tomato red.** The surrounding text continues in the parent color until the block closes.

## 8. Text Alignment within Blocks

The `.pfa-text-*` utility wraps blocks in standard directional text environments.

```
::: pfa-text-left
This block is left-aligned. The remaining width on the right is left ragged.
:::
```

This block is left-aligned. The remaining width on the right is left ragged.

```
::: pfa-text-center
This block is centered horizontally within the text column.
:::
```

This block is centered horizontally within the text column.

```
::: pfa-text-right
This block is right-aligned, flush against the right margin.
:::
```

This block is right-aligned, flush against the right margin.

## 9. Ragged Alignment (Line-Break Honoring)

The `.pfa-align-*` utility preserves and honors explicit line breaks (via backslash escapes), making it ideal for poetry, formal addresses, and titles.

```
::: pfa-align-center
First centered line\
Second centered line\
Third centered line
:::
```

First centered line  
Second centered line  
Third centered line

## 10. Shrink-to-Fit Block Alignment

The `.pfa-block-*` utility creates an isolated bounding box that shrinks dynamically to match its internal content before positioning itself in the document flow.

```
::: {.pfa-block-left}
Left-Aligned Shrink Block
Wraps tightly around its content.
:::
```

**Left-Aligned Shrink Block** Wraps tightly around its content.

```
::: {.pfa-block-center}
Centered Shrink Block
Wraps tightly around its content.
:::
```

**Centered Shrink Block** Wraps tightly around its content.

```
::: {.pfa-block-right}
Right-Aligned Shrink Block
Wraps tightly around its content.
:::
```

**Right-Aligned Shrink Block** Wraps tightly around its content.

## 11. Combining Multiple Classes

Classes compose seamlessly. Sizing, family, weight, color, and decoration definitions can stack safely on any single element.

## Inline Composition

Style	Syntax	Output
Bold Sans Display	[Sample]{.pfa-font-bold .pfa-font-sans .pfa-text-lg pfa-color="red"}	<b>Sample</b>
Italic Mono Caption	[Sample]{.pfa-font-italic .pfa-font-mono .pfa-text-sm}	<i>Sample</i>
Smallcaps Underline	[Sample]{.pfa-text-underline .pfa-font-smallcaps pfa-color="forestgreen"}	<u>SAMPLE</u>

## Block Composition

```
::: {.pfa-text-center .pfa-font-sans .pfa-font-bold .pfa-text-lg
↪ pfa-color="midnightblue"}
A centered, bold, sans-serif, large, midnight-blue block — assembled from five
↪ utilities.
:::
```

**A centered, bold, sans-serif, large, midnight-blue block —  
assembled from five utilities.**

## 12. Legacy Short-Hand Aliases (Deprecated)

**Deprecated** — retained strictly for backward compatibility. New documents should use the explicit `pfa-*` namespaces above to prevent global framework collisions.

### 12.1 Font Weight, Shape, and Family Aliases

Legacy Alias	Short	Modern Class
bold	bf	pfa-font-bold
emphasis	em	pfa-font-emphasis
italic	it	pfa-font-italic
medium	md	pfa-font-medium
monospace	tt	pfa-font-mono
normalfont	nf	pfa-font-normal
sans	sf	pfa-font-sans

Legacy Alias	Short	Modern Class
serif	rm	pfa-font-serif
slanted	sl	pfa-font-slanted
smallcaps	sc	pfa-font-smallcaps
upright	up	pfa-font-upright

## 12.2 Font Size Aliases

Legacy Alias	Modern Class
xsmall	pfa-text-xs
small	pfa-text-sm
normal	pfa-text-normal
large	pfa-text-lg
xlarge	pfa-text-xl
xxlarge	pfa-text-2xl
huge	pfa-text-3xl

## 12.3 Alignment Aliases

Legacy Alias	Modern Class
center	pfa-text-center
flushleft	pfa-text-left
flushright	pfa-text-right
centering	pfa-align-center
raggedleft	pfa-align-right
raggedright	pfa-align-left

## 12.4 Underline and Strikeout Aliases

Legacy Alias	Short	Modern Class
uline	u	pfa-text-underline
uuline	uu	pfa-text-underline-double
dashuline	dau	pfa-text-underline-dashed
dotuline	dou	pfa-text-underline-dotted
uwave	uw	pfa-text-underline-wave



Legacy Alias	Short	Modern Class
sout	so	pfa-text-strikeout
xout	xo	pfa-text-markout